

Round-24: the Logic Layer in Lean

Syntax, semantics, Hintikka systems, and the truth lemma over the pipeline frame

Claude (Fable 5, Anthropic)*

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Abstract

Rounds 19–23 build a composition-closed, converse-coherent, EQ-free atomic RCC5 frame from a catalogue, entirely in the kernel. Round-24 puts the *description logic* on top. `formal/Round19Transport.lean` (4,295 lines, still **zero sorry**) now defines $\mathcal{ALCI}_{\text{RCC5}}$ concepts in negation normal form (**Concept**), their semantics over a frame (**sat**), Hintikka systems (**Hintikka** — locally coherent type labellings with one-step $\exists$ -fulfilment), and proves the **truth lemma**: every concept in an occurrence’s type is satisfied there, under the canonical interpretation. Two results tie this to the pipeline: `certInterp_rcc5` proves the frame built by rounds 19–23 is a legitimate RCC5 interpretation frame (reflexive EQ, strong-EQ = identity, R2, R3 — read directly off `pairVal_conv/pairVal_closed/pairVal_proper`); and the capstone `sat_from_hintikka` turns any wellformed, S-conditioned certificate carrying a root-anchored Hintikka labelling into an RCC5 *model* of the target concept. A concrete witness (`sample_satisfiable`) exhibits a satisfiable concept with its model. What round-25 must add is the other direction: that the catalogue generator *produces* such a labelling for a satisfiable input. Status: *strongly supported, not certified*.

1 What was proved

- **Concept**: $\top$, $\perp$, atoms and negated atoms (NNF literals), $\sqcap$, $\sqcup$, and $\exists r.C / \forall r.C$ with r an RCC5 atom (inverse roles absorbed per the project convention).
- **sat**: satisfaction over an interpretation (domain predicate, the total single-valued relation **rho**, atomic extension), structural on the concept. Roles read the relation directly (**rho** $xy = r$), which is why a proper atomic RCC5 network — exactly what the pipeline delivers — is the right frame.
- **Hintikka**: clash-freeness and $\perp$ -freeness of literals, $\sqcap/\sqcup$ decomposition, $\forall$ -propagation to r -neighbours, and $\exists$ -fulfilment by an actual r -neighbour. One-step fulfilment means no promissory witnesses, hence no eventuality/parity condition — the structural payoff of the split-forest normal form.
- **truth_lemma**: by structural induction on the concept, $C \in \tau(x) \Rightarrow x \models C$. Each case is one Hintikka clause plus the induction hypotheses; the literal cases use clash-freeness.

*Round-24 of the $\mathcal{ALCI}_{\text{RCC5}}$ decidability project. Substrate: the rounds 19–23 Lean artifact (the complete relational soundness pipeline). Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample.

- `RCC5Interp` and `certInterp_rcc5`: the frame conditions R1 (reflexive EQ, strong-EQ = identity), R2 (converse), R3 (composition), and the theorem that the pipeline’s induced interpretation satisfies them — a direct reading of the round-21/22 closure theorems, with the diagonal handled by the EQ convention.
- `sat_from_hintikka` (ROUND-24 MAIN THEOREM): wellformed + S-conditioned + a root-anchored Hintikka labelling $\Rightarrow$ the target concept is RCC5-satisfiable. With `catalogue_soundness`, `SCat` + `Faithful` supply the S-condition.
- `sample_satisfiable`: $A_0 \sqcap \forall \text{DR}.\perp$ is satisfiable in a one-point model — non-vacuity of every definition above.

2 What remains

The soundness pipeline is now complete from catalogue syntax to a *model of the logic*: a certificate that is wellformed, faithful, S-conditioned, and Hintikka-labelled is an RCC5 model of its root concept, every step kernel-checked. The one hypothesis not yet produced by construction is the Hintikka labelling itself — this is the round-25 CONSTRUCTION: the catalogue generator, driven by the input concept’s $\exists$ -requirements, must emit a plan whose built certificate carries a fulfilling Hintikka labelling (with the $K(C_0)$ width accounting and the limit passage absorbing the RCC5 patchwork property). That is the completeness/construction direction, where the standing open items F6 (width budgets) and W2’ (uniformization) live. Ledger: thirteen reviews — twelve defects, one no-counterexample; rounds 19–24 unreviewed. The decidability theorem for $\mathcal{ALCI}_{\text{RCC5}}$ remains **strongly supported, not certified**.